

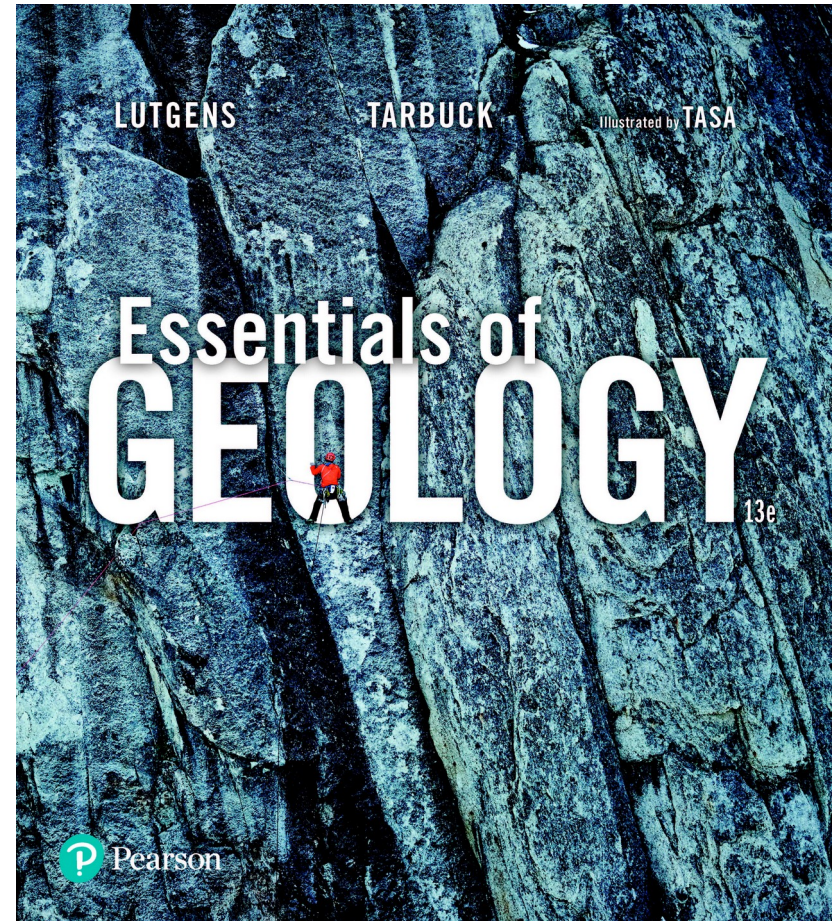
Lecture 1 Outline

An Introduction to Geology

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Essentials of Geology

Thirteenth Edition



Learning Objectives

- **LO 1.1** Distinguish between physical and historical geology and describe the connections between people and geology.
- **LO 1.2** Summarize early and modern views on how change occurs on Earth and relate them to the prevailing ideas about the age of Earth.
- **LO 1.3** Discuss the nature of scientific inquiry, including the construction of hypotheses and the development of theories.
- **LO 1.4** List and describe Earth's four major spheres. Define *system* and explain why Earth is considered to be a system.

Learning Objectives (cont.)

- **LO 1.5** Outline the stages in the formation of our solar system.
- **LO 1.6** Describe Earth's internal structure.
- **LO 1.7** Sketch, label, and explain the rock cycle.
- **LO 1.8** List and describe the major features of the continents and ocean basins.

1.1 Geology: The Science of Earth

- Geology is the science that pursues an understanding of planet Earth.
 - Physical geology examines the materials composing Earth and seeks to understand the many processes that operate beneath and upon its surface.
 - Historical geology seeks an understanding of the origin of Earth and its development through time.

1.1 Geology: The Science of Earth (cont.)

- Geology, people, and the environment
 - There are many important relationships between people and the natural environment.
 - Some of the problems and issues addressed by geology involve natural hazards, resources, world population growth, and environmental issues.

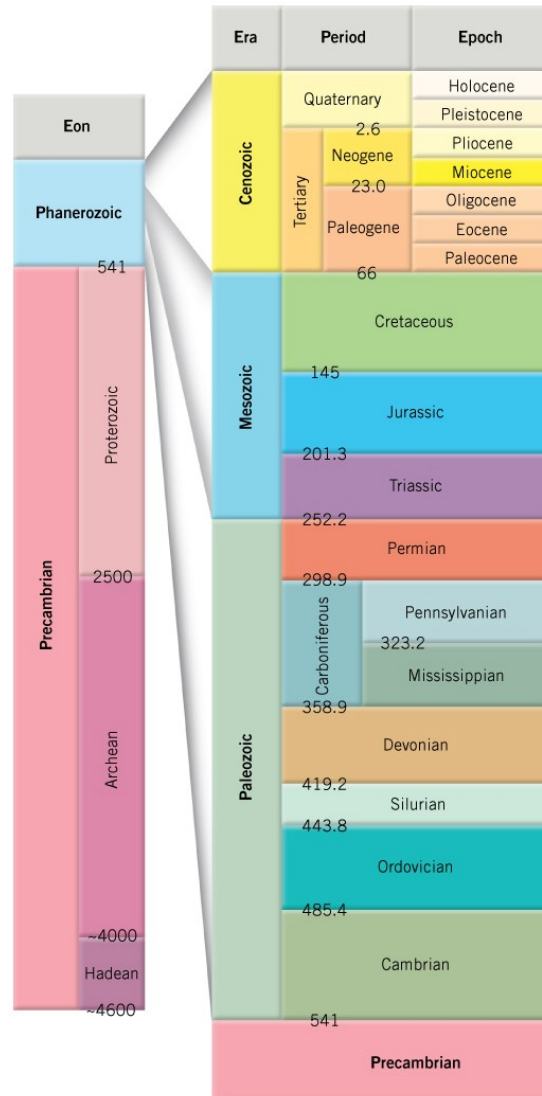
1.1 Geology: The Science of Earth (cont.)

- Some historical notes about geology
 - The nature of Earth has been a focus of study for centuries.
 - Catastrophism
 - Uniformitarianism and the birth of modern geology

1.2 The Development of Geology

- Geologists are now able to assign fairly accurate dates to events in Earth history
- Relative dating and the geologic time scale
 - Relative dating means that dates are placed in their proper sequence or order without knowing their age in years.
- The magnitude of geologic time
 - Involves vast times: Millions or billions of years
 - An appreciation for the magnitude of geologic time is important because many processes are gradual.

The Geologic Time Scale



1.3 The Nature of Scientific Inquiry

- Science assumes the natural world is consistent and predictable.
- The goal of science is to discover patterns in nature and use the knowledge to make predictions.
- Scientists collect facts through observation and measurements.

1.3 The Nature of Scientific Inquiry (cont.)

- How or why things happen are explained using a
 - Hypothesis: A tentative (or untested) explanation
 - Theory: A well-tested and widely accepted view that the scientific community agrees, best explains certain observable facts.

1.3 The Nature of Scientific Inquiry (cont.)

- Scientific method
 - Scientific method involves gathering facts through observations and formulation of hypotheses and theories.
- There is no fixed path that scientists follow that leads to scientific knowledge.

1.4 Earth as a System

- Earth is a planet that is small and self-contained.
- Earth's four spheres
 - Hydrosphere
 - Atmosphere
 - Biosphere
 - Solid Earth

A View of Earth from the Moon

[Video](#)



View called “Earthrise” that greeted *Apollo 8* astronauts as their spacecraft emerged from behind the Moon in December 1968. This classic image let people see Earth differently than ever before.

The Intersection of Three Spheres: Atmosphere, Hydrosphere, and Geosphere



1.4 Earth as a System (cont.)

- Earth is a dynamic planet with many interacting parts or spheres
- Parts of the Earth system are linked
- It is characterized by processes that
 - vary on spatial scales from fractions of a millimeter to thousands of kilometers
 - have time scales that range from milliseconds to billions of years

1.4 Earth as a System (cont.)

- The Earth system is powered by the Sun, which drives external processes
 - In the atmosphere
 - In the hydrosphere
 - At Earth's surface

1.4 Earth as a System (cont.)

- The Earth system is also powered from Earth's interior.
 - Heat remaining from the Earth's formation and heat that is continuously generated by radioactive decay powers the internal processes that produce volcanoes, earthquakes, and mountains.

1.5 Origin and Early Evolution of Earth

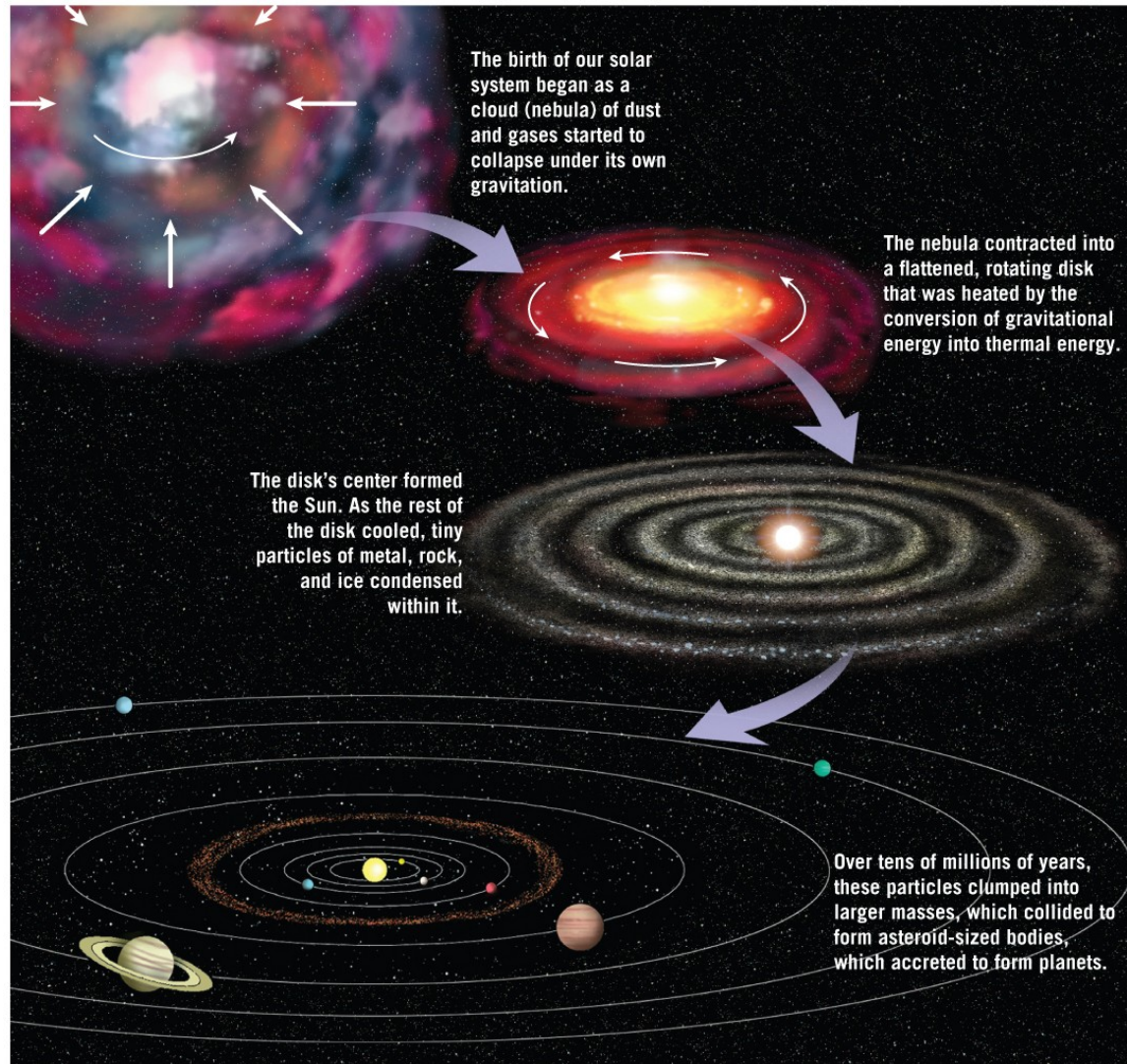
- The universe began with the Big Bang 13.7 billion years ago.
- Earth and the other planets formed at essentially the same time out of the same material as the Sun.
- The Nebular Theory proposes that the bodies of our solar system evolved from an enormous rotating cloud called the *solar nebula*.

1.5 Origin and Early Evolution of Earth (cont.)

- Nebular Theory
 - The solar nebula consisted of hydrogen and helium in addition to microscopic dust grains.
 - A disturbance caused the solar nebula to slowly collapse and rotate.
 - The solar nebula assumed a flat, disk shape with the protosun (pre-Sun) at the center.
 - Inner planets began to form from metallic and rocky substances.
 - Larger outer planets began forming from fragments of ices (H_2O , CO_2 , and others).

The Nebular Theory

Tutorial



1.5 Origin and Early Evolution of Earth (cont.)

- Formation of Earth's layered structure
 - Metals sank to the center.
 - Molten rock rose to produce a primitive crust.
- Chemical differentiation established the three basic divisions of Earth's interior (core, mantle, crust).
- A primitive atmosphere evolved from volcanic gases.
- The earliest primitive crust was lost to erosion and geologic processes.

1.6 Earth's Internal Structure

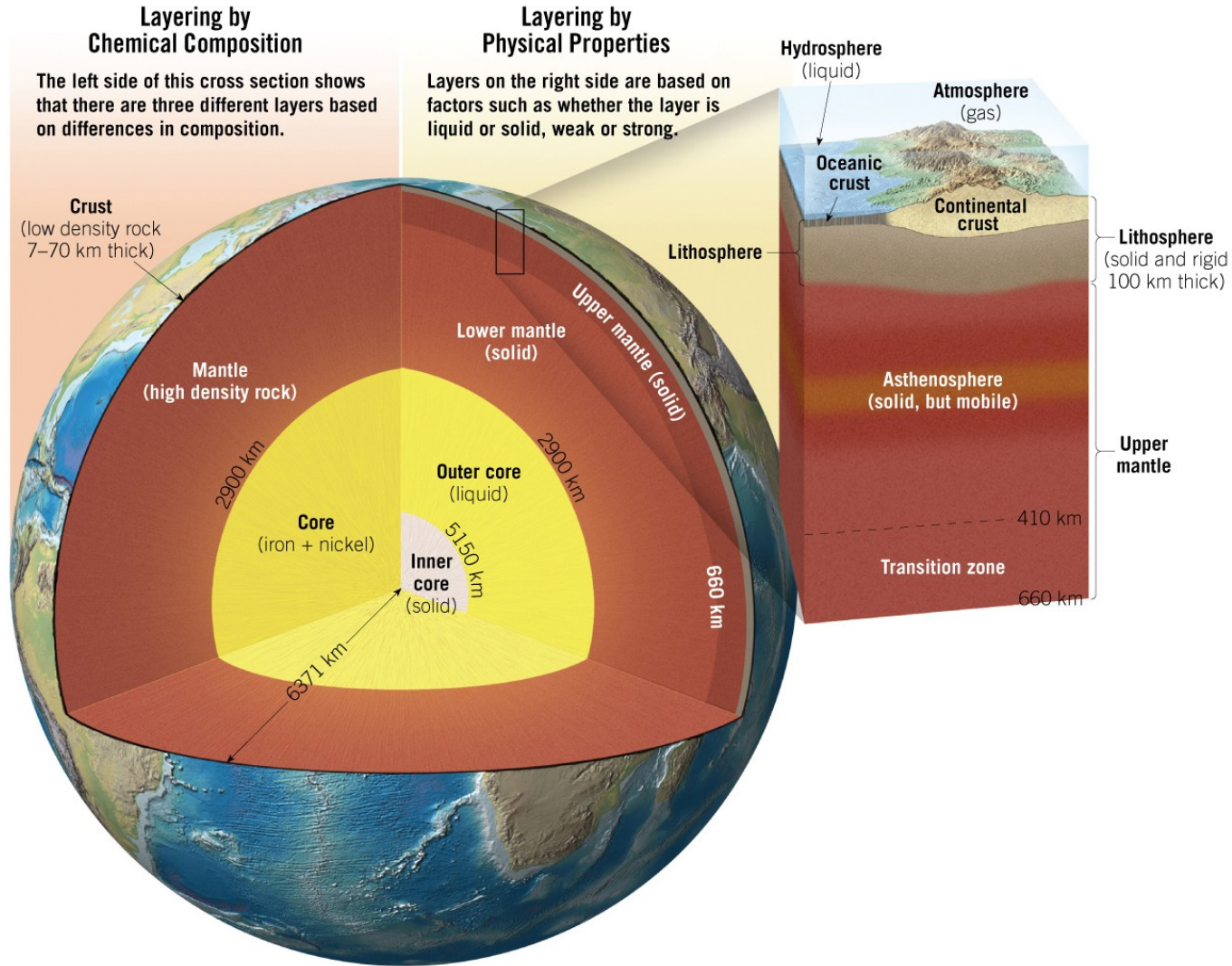
- Earth's internal layers can be defined by
 - Chemical composition
 - Physical properties
- Layers defined by composition
 - Crust
 - Mantle
 - Core

1.6 Earth's Internal Structure (cont.)

- Four main layers of Earth are based on physical properties and hence mechanical strength
 - Lithosphere
 - Asthenosphere
 - Outer core
 - Inner core

1.6 Earth's Internal Structure (cont.)

Tutorial

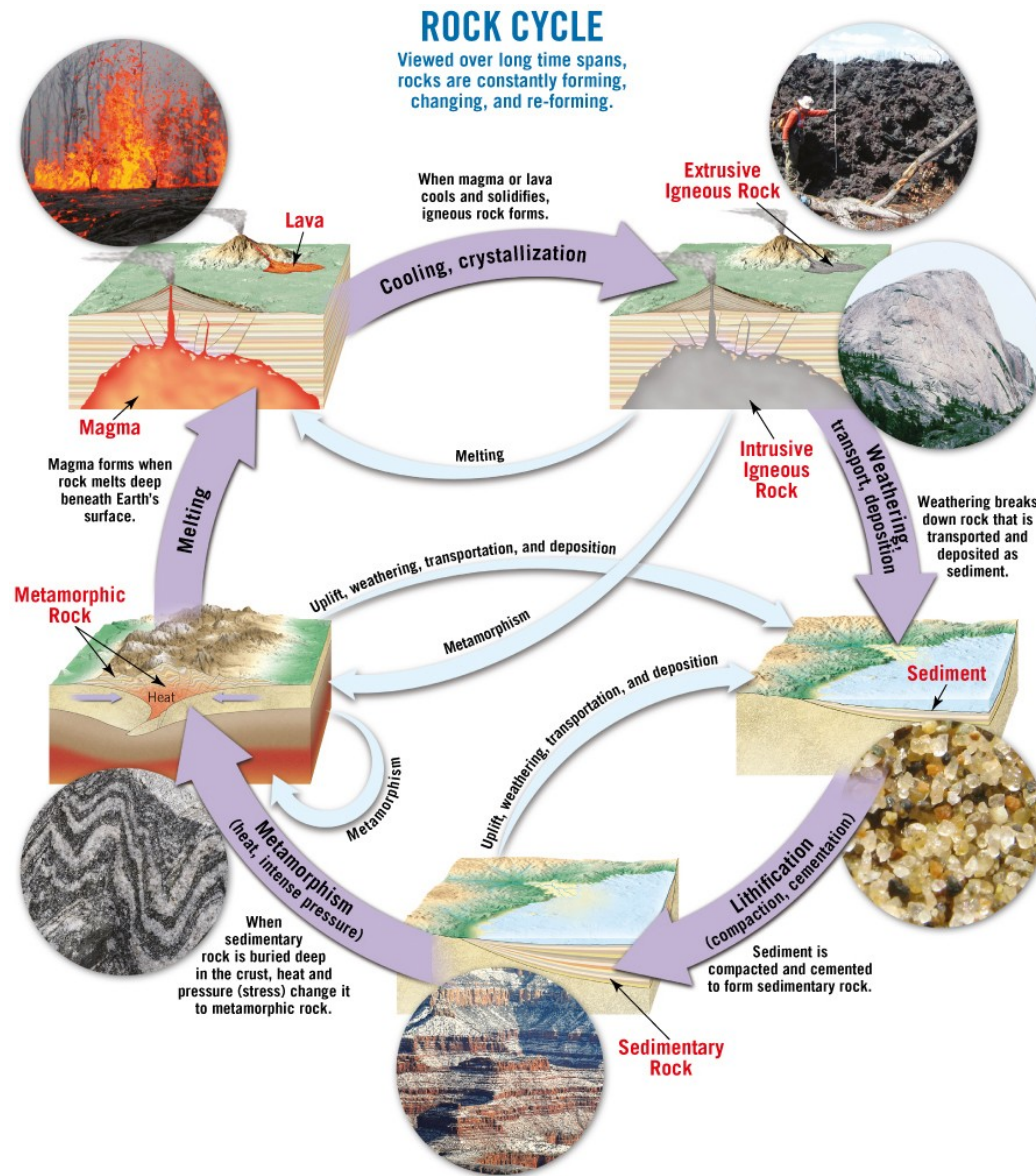


1.7 Rocks and the Rock Cycle

- Illustrates the interrelationships among different parts of the Earth system through various processes.

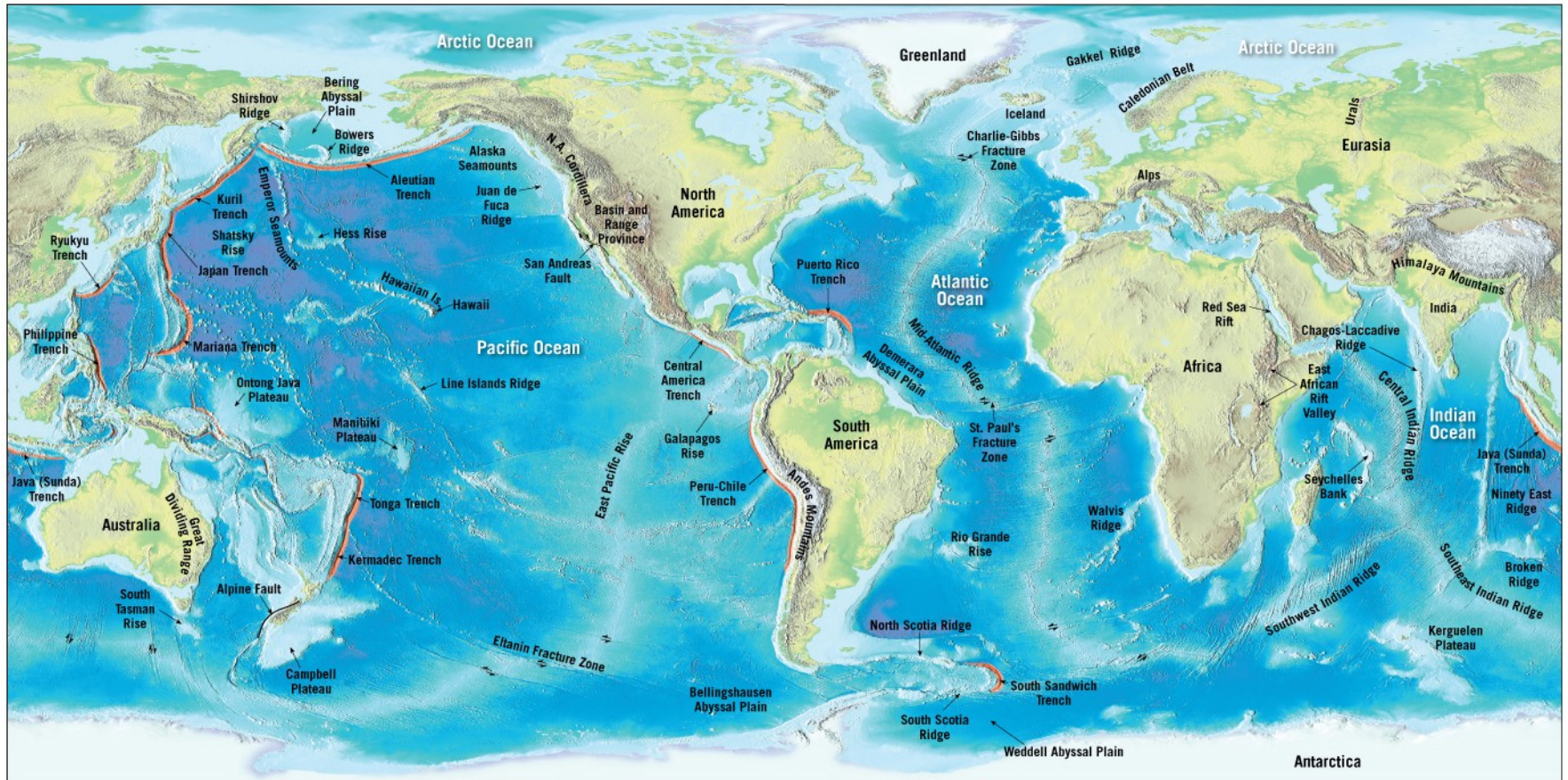
The Rock Cycle

Tutorial



1.8 The Face of Earth

- Earth's surface is divided.



1.8 The Face of Earth (cont.)

- Ocean basins
 - Continental margins
 - Includes the continental shelf, continental slope, and the continental rise
 - Deep-ocean basins
 - Abyssal plains
 - Oceanic trenches
 - Seamounts
 - Oceanic ridge system
 - Most prominent topographic feature on Earth
 - Composed of igneous rock that has been fractured and uplifted

1.8 The Face of Earth (cont.)

- Earth's surface
 - Continents
 - Oceans
- Continents
 - Mountain belts
 - Most prominent feature of continents
 - The stable interior
 - Shields and stable platforms

Earth's Mountain Belts, Stable Platforms, and Shields

Tutorial

The Canadian Shield is an expansive region of ancient Precambrian rocks, some more than 4 billion years old. It was recently scoured by Ice Age glaciers.



The Appalachians are old mountains. Mountain building began about 480 million years ago and continued for more than 200 million years. Erosion has lowered these once lofty peaks.



The rugged Himalayas are the highest mountains on Earth and are geologically young. They began forming about 50 million years ago and uplift continues today.

